

Simulation Assumption Standard - (SAS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Governed Space: Simulation Assumptions

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Canonical Language: English (UCL™)

Governed Space

Simulation Assumptions

Canonical Definition

Simulation Assumption Standard (SAS) defines the governance conditions under which assumptions necessary to construct, configure, execute, interpret, or maintain a simulation are explicitly identified, justified, bounded, related to their supporting basis, connected to their dependencies, assessed for uncertainty, constrained by their limitations, and preserved as traceable components of the simulation lifecycle.

SAS ensures that simulation behavior, outputs, and downstream conclusions cannot silently depend upon assumptions whose existence, basis, uncertainty, dependency, or consequence remains unknown.

Core Question

Which assumptions enable the simulation, and how must their basis, dependencies, uncertainty, limitations, and consequences remain governed?

Purpose

Simulation cannot operate without assumptions.

Some assumptions are explicit.

Others emerge from: incomplete knowledge, simplified reality, missing data, uncertain environments, unresolved behavior, unavailable measurements, model boundaries, computational constraints, or necessary approximations.

Assumptions may concern: physical conditions, system behavior, human behavior, autonomous behavior, environmental conditions, dependencies, parameters, distributions, operating conditions, boundary behavior, failure conditions, or future states.

The purpose of SAS is not to eliminate assumptions.

The purpose is to make them governable.

Operational Role

SAS is the fourth Parent Standard of SIMULOS® — Simulation Governance Architecture.

The preceding standards establish:

SOS — What exactly is being simulated?

SPSS — Why is it being simulated, what must it cover, and within which boundaries?

RRS — Which aspects of reality must be represented, at what abstraction, fidelity, resolution, and limitation?

SAS asks:

Which conditions must now be assumed so that this governed representation can become an executable simulation, and what risks or interpretive limits do those assumptions introduce?

SAS creates the methodological bridge between:

Governed Reality Representation

and:

Simulation Model & Environment construction.

Governance Flow

Reality Representation Basis → Assumption Need → Assumption Identification → Assumption Basis → Assumption Dependency → Assumption Uncertainty → Assumption Limitation → Assumption Consequence → Governed Simulation Assumption Basis

The resulting Simulation Assumption Basis becomes an input to:

Simulation Model & Environment Governance Simulation Assumption

A Simulation Assumption is an explicitly or implicitly adopted condition treated as sufficiently true, stable, representative, bounded, or usable for purposes of the simulation despite incomplete direct knowledge or complete representation.

An assumption may define: what is held constant, what is presumed, what is simplified, what is treated as representative, what relationship is expected to hold, what behavior is presumed, what dependency is expected to remain available, what uncertainty is bounded, or what condition is accepted for simulation purposes.

Assumption ≠ Fact

SIMULOS® establishes:

Simulation Assumption ≠ Established Fact

An assumption may be reasonable, evidence-supported, conventional, conservative, provisional, or necessary.

It must not silently become represented as independently established reality merely because the simulation depends upon it. Assumption ≠ Reality Representation

Reality Representation establishes how selected reality is represented.

Simulation Assumptions establish what must be presumed in order for that representation and subsequent simulation to function.

For example:

Represented Weather System

belongs to Reality Representation.

Weather remains within defined range X during the scenario

may be a Simulation Assumption.

The distinction must remain explicit.

Assumption ≠ Input

An assumption and an input may appear technically similar.

However:

Input provides a value, dataset, signal, or condition to simulation execution.

Assumption establishes the governing premise under which that value or condition is treated as usable or meaningful.

For example:

Input: Temperature = 25°C

Assumption: Ambient temperature remains approximately stable during the simulated event.

The two concepts should not be collapsed.

Assumption ≠ Parameter

A parameter controls or characterizes the simulation model or environment.

An assumption may determine why that parameter has a particular value, range, distribution, or behavior.

Parameterization belongs to the subsequent Model & Environment governance space.

SAS governs the assumption behind or around that parameter where relevant.

Explicit Assumption

An Explicit Simulation Assumption is formally identified and documented.

Explicit assumptions provide a stronger governance basis because they can be: reviewed, challenged, traced, tested, changed, or bounded.

Implicit Assumption

An Implicit Simulation Assumption exists where the simulation depends upon a premise that has not been formally declared.

Implicit assumptions may arise through: default settings, inherited models, software conventions, engineering habits, undocumented parameter choices, assumed independence, assumed stationarity, assumed user behavior, or unchallenged historical practices.

SAS seeks to convert materially relevant implicit assumptions into explicit governed assumptions.

Hidden Assumption

A Hidden Assumption is a materially relevant premise upon which simulation behavior depends but whose existence is not visible to downstream users or reviewers.

Hidden assumptions are methodologically dangerous because simulation outputs may appear to arise from the model alone when they actually depend upon undisclosed premises.

Assumption Identification

Assumption governance begins by asking:

What must be presumed for this simulation to operate as designed?

Potential assumptions should be sought across: Simulation Object, Scope,

Reality Representation, model behavior, environment, parameters, inputs, scenarios, dependencies, actors, interactions, timing, failure behavior, and output interpretation.

Assumption Necessity

Not every uncertain condition must become an assumption.

A condition becomes assumption-relevant where the simulation must treat it in some defined way despite incomplete certainty or direct representation.

Assumption Basis

Every material Simulation Assumption should have an identifiable Assumption Basis where possible.

The basis may derive from: observed operational data, historical behavior, empirical evidence, engineering knowledge, physical principles, scientific literature, expert judgment, design specifications, prior simulation, validated domain models, contractual conditions, regulatory requirements, conservative convention, or another documented source.

No Basis ≠ Invalid by Default

A simulation may sometimes require an assumption despite weak or unavailable supporting basis.

SAS does not prohibit this.

Instead, the assumption should be classified accordingly and its uncertainty and limitation strengthened.

Assumption Strength

Assumptions may vary in methodological strength.

Conceptually, an assumption may be:

Strongly Supported

Supported

Conditionally Supported

Weakly Supported

Unsupported but Necessary

or:

Basis Undetermined

The classification should remain tied to the simulation inquiry.

Assumption Justification

Assumption Justification explains why the simulation adopts a particular assumption.

Justification should answer:

Why is this assumption necessary?

Why is this assumption considered reasonable for this simulation?

What supports it?

Where does it stop being reasonable?

Assumption Materiality

A Simulation Assumption is material where changing, violating, or removing the assumption could materially affect: simulation behavior, scenario progression, simulation outputs, interpretation, comparison, reality correlation, downstream evidence, or reliance.

Material assumptions require stronger governance than assumptions with negligible effect.

Materiality Is Purpose-Relative

The same assumption may be material in one simulation and immaterial in another.

Example:

Stable network latency

may be immaterial for:

high-level strategic simulation

but critical for:

real-time autonomous coordination simulation.

Critical Simulation Assumption

A Critical Simulation Assumption is an assumption whose failure could materially alter or invalidate a major simulation conclusion.

Critical assumptions should remain highly visible.

They may require: stronger basis, stronger uncertainty treatment, explicit scenario testing, sensitivity analysis, tighter monitoring, or narrower interpretation.

Assumption Hierarchy

Assumptions may exist at different levels:

Foundational Assumptions

support the overall simulation framework.

Structural Assumptions

support model architecture or relationships.

Environmental Assumptions

define simulated surrounding conditions.

Behavioral Assumptions

define actor or system behavior.

Parameter Assumptions

support selected parameter values or ranges.

Scenario Assumptions

support specific simulated conditions.

Interpretive Assumptions

affect how outputs are understood.

SAS allows such classification without requiring one universal taxonomy.

Physical Assumption

A Physical Assumption concerns a physical relationship, condition, or constraint.

Examples include: constant gravity, material stability, friction behavior, heat transfer behavior, or known physical boundary conditions.

Behavioral Assumption

A Behavioral Assumption concerns expected behavior of: humans, AI systems, autonomous agents, organizations, biological systems, or other actors.

Behavioral assumptions may be especially sensitive where behavior is adaptive, stochastic, strategic, or context-dependent.

Environmental Assumption

An Environmental Assumption concerns external conditions within the simulation environment.

Examples may include: weather range, infrastructure availability, traffic behavior, communication availability, resource availability, or environmental stability.

Dependency Assumption

A Dependency Assumption concerns the availability, behavior, reliability, or characteristics of something upon which the Simulation Object or simulation environment depends.

Examples include: network availability, external database access, navigation service availability, energy supply, human supervision, external AI services, or supporting infrastructure.

Independence Assumption

Simulations frequently assume that variables, actors, events, or failures are independent.

This may dramatically simplify simulation.

Where independence is false, resulting outputs may be materially distorted.

Therefore:

Assumed Independence must remain explicit where dependence could materially change simulation behavior.

Stationarity Assumption

A Stationarity Assumption treats a relevant process or distribution as sufficiently stable over time.

This may concern: human behavior, failure rates, demand, environmental conditions, financial relationships, system performance, or operating conditions.

Dynamic systems may invalidate stationarity assumptions rapidly.

Distribution Assumption

Stochastic simulations often rely upon assumed probability distributions.

The chosen distribution should remain traceable to: evidence, domain knowledge, prior observations, theoretical basis, or explicit approximation.

Boundary Assumption

A Boundary Assumption concerns conditions at or beyond the simulation boundary.

For example:

External network behaves according to defined latency profile.

The network may remain outside the Simulation Object while its behavior is represented through an assumption.

Failure Assumption

A simulation may require assumptions regarding: failure frequency, failure independence, propagation, degradation, recovery, or failure severity.

Where safety or resilience is material, such assumptions should receive explicit governance.

Human Behavior Assumption

Human behavior may require assumptions concerning: reaction time, decision-making, compliance, intervention, error, variability, coordination, or supervision.

SAS governs the assumption role.

CLIA® may govern the underlying cognition where applicable.

AI & Autonomous Behavior Assumption

Simulation of AI or autonomous systems may depend upon assumptions about: behavior stability, tool availability, permissions, perception, adaptation, coordination, escalation, or environmental response.

AIG® and ASGA™ govern the underlying AI or autonomous operational spaces.

SAS governs specifically assumptions used inside the simulation.

Assumption Dependency

Assumptions may depend upon: other assumptions, evidence, model structure, parameters, environmental conditions, external systems, scenario conditions, or operational constraints.

The relationship should remain visible where the validity or failure of one assumption can affect another.

Assumption Chain

Simulation assumptions may form chains:

Assumption A → Assumption B → Parameter Choice → Scenario Behavior → Simulation Output

A weakness upstream may therefore propagate into downstream results.

SAS requires material assumption chains to remain traceable.

Assumption Interaction

Two individually reasonable assumptions may interact to produce an unreasonable combined simulation condition.

For example:

Assumption A: Human operator responds within expected range

and:

Assumption B: Network latency remains within expected range

may individually be reasonable.

Their simultaneous worst-case conditions may create materially different system behavior.

SAS therefore allows assumption interaction to be governed collectively.

Assumption Accumulation

A simulation may contain hundreds or thousands of assumptions.

Individually small assumptions may accumulate into a material combined effect.

SAS establishes:

Assumption materiality must be considered both individually and collectively where cumulative effects may alter simulation meaning.

Assumption Uncertainty

An assumption inherently contains some degree of uncertainty unless it is effectively an established condition.

SAS requires material uncertainty to remain visible.

Uncertainty may concern: whether the assumption is true, how broadly it applies, how long it remains true, how stable it is, how variable the assumed condition may be, or how sensitive the simulation is to assumption failure.

Assumption Uncertainty ≠ General Simulation Uncertainty

SAS governs uncertainty specifically attached to Simulation Assumptions.

It does not own every uncertainty in the simulation lifecycle.

Other uncertainty may arise from: data, model structure, execution, measurement, representation, scenarios, or operational reality.

Assumption Range

Where a single assumed value is unjustified, an assumption may require a range.

For example:

Reaction Time = 0.8–1.5 seconds

may be more methodologically appropriate than:

Reaction Time = 1.0 second

where real variability is material.

Probabilistic Assumption

An assumption may be expressed probabilistically.

This may be appropriate where uncertainty or variability can be represented as: distribution, confidence interval, probability, likelihood class, or stochastic rule.

Conditional Assumption

A Conditional Assumption applies only while a defined condition remains true.

For example:

Navigation service assumed available while satellite visibility remains above threshold X.

When the condition changes, the assumption may no longer be applicable.

Scenario-Specific Assumption

Some assumptions belong only to particular scenarios.

A scenario-specific assumption must not automatically become a global simulation assumption.

Assumption Sensitivity

An Assumption Sensitivity Assessment asks:

How much does simulation behavior or output change when this assumption changes within a plausible range?

Highly sensitive assumptions may require: stronger basis, wider scenario exploration, explicit uncertainty propagation, or narrower output interpretation.

Assumption Stress

A simulation may intentionally test what happens when an assumption is weakened, violated, or removed.

This helps distinguish:

Assumption-dependent result

from:

robust simulation result.

Assumption Failure

An Assumption Failure occurs where an assumption becomes materially inconsistent with the represented or observed condition.

This may occur: during simulation, after new data appears, when reality changes, when a model changes, when an environment changes, or when new evidence demonstrates the assumption was incorrect.

Assumption Failure Does Not Automatically Invalidate Entire Simulation

The consequence depends upon: materiality, affected scope, dependency chain, scenario, outputs, and whether the assumption can be replaced or bounded.

SAS therefore requires consequence assessment rather than automatic total invalidation.

Assumption Consequence

A material assumption should identify what could happen if it is wrong or materially violated.

Possible consequences include:

No Material Effect

Local Simulation Effect

Scenario Limitation

Output Qualification

Model Reconfiguration Required

Additional Simulation Required

Reality Representation Reassessment Required

Operational Interpretation Restricted

or:

Consequence Undetermined Assumption Limitation

An Assumption Limitation identifies the boundary beyond which the assumption should not support simulation interpretation.

For example:

Assumption valid only for temperatures between X and Y

or:

Assumption applies only to normal operating conditions.

Assumption Applicability

Each material assumption should remain connected to where it applies.

Applicability may be bounded by: Simulation Object version, configuration, environment, scenario, operating mode, population, temporal period, spatial region, or other simulation context.

Assumption Status

A Simulation Assumption may hold a status such as:

Established

Conditionally Established

Weakly Supported

Materially Uncertain

Conflicted

Failed

Superseded

or:

Undetermined

Status does not establish universal truth.

It establishes the current governance condition of the assumption within the simulation.

Assumption Conflict

An Assumption Conflict exists where two assumptions or assumption bases materially contradict one another.

Example:

Assumption A: Communication delay is negligible

and:

Assumption B: Control behavior depends upon material network latency

cannot both remain globally true without additional contextual separation.

Material conflicts should be resolved or preserved explicitly.

Assumption Consistency

Assumptions across interacting simulation components should remain mutually coherent.

One component should not operate under:

constant environment

while another depends upon:

rapidly changing environment

without explicit governance explaining the difference.

Assumption Completeness

Assumption Completeness asks whether material assumptions supporting the simulation have been sufficiently identified.

It does not require documenting trivial mathematical conventions with no material simulation consequence.

It requires that material premises do not remain hidden.

Assumption Register

A governed simulation should maintain a persistent Simulation Assumption Register where appropriate.

The register may include: Assumption Identifier, assumption statement, assumption type, affected object or process, basis, justification, dependencies, uncertainty, materiality, sensitivity, applicability, limitations, consequences, status, change history, and downstream references.

No Hidden Assumption Rule

SIMULOS® establishes:

A material simulation conclusion must not depend upon a material assumption that is absent from the governed simulation record.

No Assumption-to-Fact Conversion Rule

Repeated use of an assumption does not transform it into an established fact.

No Convenience Justification Rule

An assumption must not be considered methodologically justified merely because: data are unavailable, modeling is difficult, computation is expensive, the software uses that default, or the assumption produces convenient results.

These may explain why the assumption exists.

They do not establish its methodological strength.

No Assumption Laundering Rule

An assumption introduced upstream must not disappear downstream and later be represented as observed or validated reality.

Conceptually:

Assumption → Model → Output

must not silently become:

Output → Evidence of the Assumption

without an independent basis.

No Circular Assumption Confirmation Rule

A simulation should not be used as sole proof that one of its own foundational assumptions is correct where the output structurally depends upon that assumption.

No Assumption Stacking Concealment Rule

Several weak assumptions must not be treated as one strong simulation basis merely because their combined role is hidden.

No Assumption Scope Expansion Rule

An assumption valid within one context must not silently become applicable outside that context.

No Retrospective Assumption Redefinition Rule

Assumptions should not be materially redefined after outputs are known merely to make the simulation appear successful.

Conservative Assumption

A simulation may intentionally adopt conservative assumptions.

This can be appropriate in high-consequence settings.

However:

Conservative ≠ Automatically Valid

The conservative direction, basis, and consequence should remain explicit.

Optimistic Assumption

Optimistic assumptions may also be legitimate in exploratory simulations.

Their role should remain visible so that optimistic assumptions are not later treated as neutral operational conditions.

Worst-Case Assumption

A worst-case assumption may support: stress testing, safety analysis, contingency planning, or resilience analysis.

The assumed condition should be distinguishable from expected operational reality.

Representative Assumption

A simulation may use a representative value or behavior.

The chosen representation should identify:

representative of what population, condition, environment, or period?

Without such context, “representative” can become meaningless.

Assumption Change

Simulation Assumptions may change when: new evidence emerges, Operational Reality changes, the Simulation Object changes, Reality Representation changes, new scenarios are added, model structure changes, parameterization changes, or previous assumptions fail.

Material changes should remain explicitly traceable.

Assumption Supersession

A new assumption may supersede a previous assumption.

The previous assumption should remain historically traceable where it affected earlier simulation activity.

Assumption Change Propagation

A material Assumption Change may affect: Simulation Model & Environment, parameters, Inputs & Data, Scenario Design, Execution, outputs, Reality Correlation, Simulation Evidence, and downstream reliance.

SAS therefore treats assumptions as active lifecycle dependencies rather than static documentation.

Assumption Reassessment Trigger

A material change in: evidence, reality, object version, configuration, environment, model, dependency, population, scenario, or intended use

may generate an Assumption Reassessment Trigger.

Assumption Versioning

Where a material assumption changes, the relevant assumption state should remain versioned or historically distinguishable.

Simulation runs based upon materially different assumptions should not automatically be treated as directly equivalent.

Assumption-to-Execution Binding

Simulation execution should remain traceable to the set of material assumptions active during that run.

Conceptually:

Simulation Assumption Set → Simulation Model & Environment → Simulation Execution → Simulation Output

This allows later reconstruction of why outputs differed between simulation runs.

Assumption-to-Output Binding

Where a material output depends strongly upon one or more assumptions, that dependency should remain traceable.

This is especially important for: high-consequence conclusions, sensitivity-sensitive outputs, extrapolated results, uncertain environments, or simulations used as validation evidence.

Assumption and Reality Correlation

Later Reality Correlation may demonstrate that an assumption corresponds poorly to observed reality.

This does not mean SAS performs correlation itself.

It means the assumption lifecycle must remain capable of receiving new information from downstream reality comparison.

Assumption and Validation

SAS does not determine whether an assumption is universally valid.

VALIDOS® may validate: the assumption itself, its evidence basis, the simulation, the resulting claim, or another relevant Validation Object.

The boundary is:

SAS governs the assumption as part of simulation.

VALIDOS® governs formal validation where required.

Assumption and Operational Reality

ORA™ may establish actual operational conditions relevant to an assumption.

SAS does not replace ORA™.

It governs how simulation treats incomplete, expected, bounded, or presumed conditions relative to that reality.

Assumption and Integrity

INTEGROS® may govern integrity of: assumption records, source records, configuration, traceability, or related lifecycle artifacts.

SAS governs specifically assumption identification, basis, dependency, uncertainty, limitation, and simulation consequence.

Assumption and Auditability

ADIT® may preserve evidence showing: when assumptions were introduced, how they changed, which simulation executions used them, and what outputs depended upon them.

Assumption and Risk

RIS™ may inform which assumptions require increased attention where assumption failure creates high consequence.

SAS does not become a general risk architecture.

Assumption Basis Record

A governed Simulation Assumption Basis should preserve enough information to reconstruct: which assumptions existed, why they existed, what supported them, what depended upon them, how uncertain they were, where they applied, what

limitations existed, and what consequences could result from failure.

Operational Space

SAS governs: assumption identification, assumption basis, assumption justification, assumption classification, assumption materiality, critical assumptions, assumption dependencies, assumption chains, assumption interaction, assumption accumulation, assumption uncertainty, assumption ranges, conditional assumptions, scenario-specific assumptions, assumption sensitivity, assumption failure, assumption consequences, assumption limitations, assumption applicability, assumption status, assumption conflict, assumption completeness, assumption changes, assumption supersession, assumption reassessment, and assumption traceability.

Governance Boundary

SAS begins when:

a governed Reality Representation Basis exists and the simulation requires premises, approximations, expected conditions, or unresolved relationships to proceed into model and environment construction.

SAS ends when:

material Simulation Assumptions have been sufficiently identified, supported or explicitly qualified, linked to dependencies, bounded by uncertainty and limitations, connected to their consequences, and preserved as a governed Simulation Assumption Basis suitable for Simulation Model & Environment governance.

Governance Scope

SAS governs assumptions specifically within the simulation lifecycle.

It does not govern: universal truth determination, formal assumption validation, general evidence validity, actual Operational Reality, model construction itself, simulation environment configuration, parameter implementation, input-data fitness, scenario execution, output validation, general integrity, risk governance, or operational reliance authorization.

These responsibilities remain within the appropriate SIMULOS® Parent Standards or complementary OOF® architectures.

Minimum Governance Requirements

A conforming application of SAS should establish:

1. identifiable material Simulation Assumptions, 2. explicit conversion of materially relevant implicit assumptions where feasible, 3. the basis and justification for material assumptions, 4. assumption classification and materiality, 5. identification of critical assumptions, 6. material assumption dependencies, 7. assumption chains and interactions where relevant, 8.

uncertainty associated with material assumptions, 9. applicability and conditionality, 10. sensitivity where assumption variation could materially affect outputs, 11. explicit assumption limitations, 12. consequences of assumption failure where material, 13. assumption conflict handling, 14. traceability of material Assumption Changes, 15. and a persistent Simulation Assumption Basis.

Governance Outputs

SAS may produce: Simulation Assumption Register, Simulation Assumption Record, Assumption Identification Record, Implicit Assumption Record, Hidden Assumption Finding, Assumption Basis Record, Assumption Justification Record, Assumption Classification, Assumption Materiality Assessment, Critical Simulation Assumption Record, Assumption Hierarchy, Physical Assumption Record, Behavioral Assumption Record, Environmental Assumption Record, Dependency Assumption Record, Independence Assumption Record, Stationarity Assumption Record, Distribution Assumption Record, Boundary Assumption Record, Failure Assumption Record, Assumption Dependency Map, Assumption Chain, Assumption Interaction Assessment, Assumption Accumulation Assessment, Assumption Uncertainty Record, Assumption Range, Probabilistic Assumption Record, Conditional Assumption Record, Scenario-Specific Assumption Record, Assumption Sensitivity Assessment, Assumption Failure Record, Assumption Consequence Record, Assumption Limitation Record, Assumption Applicability Record, Assumption Status, Assumption Conflict Record, Assumption Completeness Assessment, Assumption Change Record, Assumption Supersession Record, Assumption Reassessment Trigger, Assumption Version Record, Assumption-to-Execution Binding, Assumption-to-Output Map, Simulation Assumption Traceability Map, and Governed Simulation Assumption Basis.

Operational Applications

Simulation Assumption Standard may be applied across: artificial intelligence, autonomous systems, robotics, aerospace, aviation, spacecraft and mission simulation, autonomous vehicles, digital twins, industrial systems, manufacturing, scientific modeling, environmental simulation, healthcare simulation, financial simulation, infrastructure simulation, energy systems, population simulation, behavioral simulation, organizational simulation, system-of-systems simulation, training environments, safety analysis, failure analysis, and future operational ecosystems.

Relationship to Other OOF® Architectures

Simulation Assumption Standard interoperates with: GOA™ ORA™ VALIDOS® INTEGROS® ADIT® AGA™ AIG® CLIA® MGIA™ ASGA™ RIS™

while preserving strict architectural boundaries.

ORA™ may establish the actual Operational Reality against which simulation assumptions are eventually compared.

VALIDOS® may formally validate assumptions, their evidence basis, simulation outputs, or the simulation itself where validation is required.

INTEGROS® may govern integrity of assumption records and associated simulation

artifacts.

ADIT® may preserve the historical evidence and traceability of assumption creation and change.

AGA™ may govern accountability for material assumption decisions.

RIS™ may influence the governance intensity required where Assumption Failure creates substantial consequence.

AIG®, CLIA®, MGIA™, and ASGA™ may define domain-specific characteristics relevant to assumptions about AI behavior, cognition, memory, and autonomous operation.

SAS consumes those characteristics where necessary without replacing their governance spaces.

Architectural Position

Simulation Assumption Standard (SAS) is the fourth Parent Standard of SIMULOS®
— Simulation Governance Architecture.

The architecture progresses:

Simulation Object

↓

Simulation Purpose & Scope

↓

Reality Representation

↓

Simulation Assumptions

SOS establishes:

What exactly is being simulated?

SPSS establishes:

Why is it being simulated and what must the inquiry cover?

RRS establishes:

Which aspects of reality must be represented and how?

SAS establishes:

Which premises must be accepted so that this representation can become an operational simulation, and what uncertainty, dependency, and limitation do those premises introduce?

Once the governed Simulation Assumption Basis exists, the next architectural question becomes:

How must the simulation model and simulation environment be constructed, configured, bounded, parameterized, and version-controlled so that the governed object, representation, and assumptions become an executable simulation system?

This transition leads directly into:

Simulation Model & Environment Standard (SMES).

Foundational Principle

A simulation may rely upon assumptions, but no material simulation conclusion should become stronger than the assumptions upon which that conclusion materially depends.

Canonical Closing Statement

Simulation Assumption Standard (SAS) establishes the assumption-governance layer of SIMULOS® by making the premises necessary for simulation explicit, attributable, bounded, uncertainty-aware, dependency-aware, and traceable throughout the simulation lifecycle. Through governance of assumption identification, basis, justification, materiality, dependencies, interaction, uncertainty, sensitivity, applicability, failure, limitations, consequences, change, and output dependence, SAS prevents simulation assumptions from becoming invisible facts merely because they are embedded within models or repeatedly reused. By establishing a governed Simulation Assumption Basis between Reality Representation and Simulation Model & Environment construction, SAS ensures that simulation behavior and outputs remain inseparable from the premises that materially shape them.

Module Architecture

→ [Simulation Assumption Identification Module \(SAIM\)](#)

→ [Simulation Assumption Basis Module \(SABM\)](#)

→ [Simulation Assumption Dependency Module \(SADM\)](#)

→ [Simulation Assumption Uncertainty Module \(SAUM\)](#)

→ [Simulation Assumption Limitation Module \(SALM\)](#)

Related Documents

→ [About Simulation Assumption Standard \(SAS\)](#)

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Canonical Source

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